

AI Report

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\section{Introduction} - 6/22/2026

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Digital streaming completely reshaped the music business. Global revenues hit USD 29.6 billion in 2024, reflecting massive growth in the streaming era~\cite{ifpi2024report}. But for independent musicians, this economic boom often feels like an illusion. I have spent 14 years playing drums in various bands across Colombia, and the reality on the ground is stark: new artists simply do not get recommended. It is not because their music lacks quality. It is because algorithms systematically favor established acts.

Most platforms still lean heavily on collaborative filtering. They just look at what people already click on. It works fine for global superstars, but it creates a brutal cold-start problem for anyone else~\cite{schein2002methods,adomavicius2005toward}. If a new track has no play history, the algorithm ignores it. This creates a vicious feedback loop. In Latin American music scenes, the situation is even worse. Local artists are not just competing with global hits; they are drowning in a sea of algorithmically-generated background music.

Content-based approaches offer a way out. By looking at the actual lyrics, acoustic properties, and metadata of a song, we can recommend based on what the music actually sounds like~\cite{ricci2015recommender}. Transformers~\cite{vaswani2017attention} are particularly good at this. Their ability to map long-range dependencies makes them ideal for capturing the complex structure of musical content~\cite{baltruvsaitis2018multimodal}.

This paper introduces EchoPulse AI. We built it as a locally-deployed multimodal recommender within the IMS Research Group at Fundación Universitaria de Popayán. Why local? Simple. We had a limited budget for cloud services, and internet connectivity in our region is notoriously unstable. Instead of seeing these constraints as a weakness, we used them to force a design that prioritizes efficiency, reproducibility, and zero cloud dependency.

We make three main contributions. First, we propose a hybrid local data architecture. It follows a five-stage paradigm to merge messy, heterogeneous music datasets without needing AWS or GCP. Second, we design a Triple-Encoder Transformer. It processes lyrics, audio, and conversational data through specialized modules,

backed by strict mathematical formulations. Third, we provide an end-to-end evaluation. We prove that you can drastically improve diversity and surface emerging artists without tanking your accuracy metrics.

Building the Master Table was easily the hardest part of this project. We had to reconcile completely different ID systems, handle massive amounts of missing data, and force four distinct datasets into a single, clean collection of 5,710 tracks. That data engineering struggle directly shaped the architecture you will read about next.

We structure the rest of the paper to walk through our approach step-by-step. Section~\ref{sec:related} covers the background on Transformers and local data architectures. Section~\ref{sec:methodology} dives into the five-stage pipeline and the model math. We detail the hardware setup and metrics in Section~\ref{sec:evaluation}, followed by an honest look at our limitations in Section~\ref{sec:discussion}. Finally, Section~\ref{sec:conclusion} wraps up the findings and what they mean for the future of equitable music recommendation.

● Sentences that are likely AI-generated.

FAQs

What is GPTZero?

GPTZero is the leading AI detector for checking whether a document was written by a large language model such as ChatGPT. GPTZero detects AI on sentence, paragraph, and document level. Our model was trained on a large, diverse corpus of human-written and AI-generated text with support for English, Spanish, French, German, and other languages. To date, GPTZero has served over 17 million users around the world, and works with over 100 organizations in education, hiring, publishing, legal, and more.

When should I use GPTZero?

Our users have seen the use of AI-generated text proliferate into education, certification, hiring and recruitment, social writing platforms, disinformation, and beyond. We've created GPTZero as a tool to highlight the possible use of AI in writing text. In particular, we focus on classifying AI use in prose. Overall, our classifier is intended to be used to flag situations in which a conversation can be started (for example, between educators and students) to drive further inquiry and spread awareness of the risks of using AI in written work.

Does GPTZero only detect ChatGPT outputs?

No, GPTZero works robustly across a range of AI language models, including but not limited to ChatGPT, GPT-5, GPT-4, GPT-3, Gemini, Claude, and AI services based on those models.

What are the limitations of the classifier?

The nature of AI-generated content is changing constantly. As such, these results should not be used to punish students. We recommend educators to use our behind-the-scenes [Writing Reports](#) as part of a holistic assessment of student work. There always exist edge cases with both instances where AI is classified as human, and human is classified as AI. Instead, we recommend educators take approaches that give students the opportunity to demonstrate their understanding in a controlled environment and craft assignments that cannot be solved with AI. Our classifier is not trained to identify AI-generated text after it has been heavily modified after generation (although we estimate this is a minority of the uses for AI-generation at the moment). Currently, our classifier can sometimes flag other machine-generated or highly procedural text as AI-generated, and as such, should be used on more descriptive portions of text.

I'm an educator who has found AI-generated text by my students. What do I do?

Firstly, at GPTZero, we don't believe that any AI detector is perfect. There always exist edge cases with both instances where AI is classified as human, and human is classified as AI. Nonetheless, we recommend that educators can do the following when they get a positive detection: Ask students to demonstrate their understanding in a controlled environment, whether that is through an in-person assessment, or through an editor that can track their edit history (for instance, using our [Writing Reports](#) through Google Docs). Check out our list of [several recommendations](#) on types of assignments that are difficult to solve with AI.

Ask the student if they can produce artifacts of their writing process, whether it is drafts, revision histories, or brainstorming notes. For example, if the editor they used to write the text has an edit history (such as Google Docs), and it was typed out with several edits over a reasonable period of time, it is likely the student work is authentic. You can use GPTZero's Writing Reports to replay the student's writing process, and view signals that indicate the authenticity of the work.

See if there is a history of AI-generated text in the student's work. We recommend looking for a long-term pattern of AI use, as opposed to a single instance, in order to determine whether the student is using AI.